

DEBIDUTTA BEHERA

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience designing, developing, and testing backend services and full-stack applications using Python, Django, Flask, FastAPI, REST APIs, and SQL/NoSQL databases. Proven track record at DRDO's Integrated Test Range achieving 99.8% system uptime, reducing deployment errors by 25%, and improving execution performance by 35% through proactive optimization and automated testing (Selenium, PyTest). Strong foundation in object-oriented programming, software development lifecycle (SDLC) practices, version control (Git/GitHub), and cross-functional collaboration. Seeker mindset with a consistent focus on identifying and delivering optimized, scalable, and reliable software solutions.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), TypeScript, JavaScript, SQL, PHP, C++, C#

Frameworks & APIs: Django, Django REST Framework, Flask, FastAPI, Node.js, Express.js, REST API Design, OOP, OR-Tools CP-SAT

Frontend: React.js, Next.js, Angular (TypeScript), HTML5, CSS3, Responsive Design

Databases: MySQL, PostgreSQL, MongoDB, SQLite, Railway — schema design, joins, aggregations, indexing

Testing & QA: Selenium, PyTest, Cross-Browser Testing, Automated HTML Test Reports, Debugging

Tools & Practices: Git, GitHub, Postman, VS Code, PyCharm, CI/CD Principles, Linux, GCP (Familiarity), Agile/SDLC

PROFESSIONAL EXPERIENCE

Software Engineer Apprentice

Jan 2025 – Jan 2026

Integrated Test Range (ITR), DRDO — Chandipur, Odisha, India | Contract No.: CORBACGE24011993125

- Engineered and maintained Python-based backend services and modules for mission-critical enterprise software, achieving 99.8% system uptime through robust design and full lifecycle ownership.
- Automated software testing pipelines using Python, Selenium, and PyTest, reducing manual testing effort by 40% and deployment defects by 25%.
- Profiled and optimized application code, eliminating runtime bottlenecks and achieving a 35% reduction in execution time.
- Contributed across the full software development lifecycle — requirement analysis, design, implementation, testing, deployment, and documentation.
- Participated in peer code reviews, authored technical documentation, and collaborated cross-functionally, consistently identifying and driving process improvements.

Full Stack Development Training Program

2024

JSpiders – Technology Training Institute, Bangalore, India

- Completed an intensive, project-based training program covering Python, Django, SQL, REST API development, and JavaScript fundamentals.
- Built multiple hands-on applications applying object-oriented programming, database design, and full-stack development best practices, establishing the technical foundation later applied during the DRDO apprenticeship.

KEY PROJECTS

Cinema Seat Reservation System — Python, Django REST Framework, React.js, TypeScript, PostgreSQL

- Architected RESTful backend endpoints with Django REST Framework to handle real-time booking transactions, seat-tier pricing logic, and data persistence in PostgreSQL.
- Designed a normalized PostgreSQL schema for users, bookings, and pricing tiers, optimizing query performance for concurrent transactions.
- Built a component-driven React.js/TypeScript frontend with real-time state management and responsive rendering.

AWS Agreement RAG Chatbot — *Python, FastAPI, Streamlit, LangChain, SQL*

- Developed an AI-powered RAG chatbot for natural-language querying of AWS agreement documents, with a FastAPI backend for document processing, embeddings, and retrieval-based QA.
- Integrated LangChain and vector search to deliver accurate, context-aware responses.

Digital Attendance Tracker — *Python, Django, React.js, TypeScript, PostgreSQL, MySQL*

- Developed a full-stack web portal with role-based access control for administrators and employees using Django and React.js/TypeScript.
- Implemented secure authentication and a normalized PostgreSQL/MySQL schema, reducing manual attendance errors and streamlining HR operations.

Automated Software Testing Framework — *Python, Selenium, PyTest*

- Engineered a cross-browser automated testing framework validating login workflows, user interactions, and form functionality across Chrome, Firefox, and Edge.
- Generated structured HTML test reports with automated screenshot capture on failure, reducing average debugging time and enabling faster CI/CD feedback loops.

E-Commerce Frontend Web Application — *React.js, TypeScript, HTML5, CSS3*

- Built an e-commerce front-end application with product listing, cart, and checkout workflows using component-based architecture.
- Implemented API integration with async/await for product data fetching, ensuring cross-browser compatibility.

AI-Powered Personal Assistant (JARVIS) — *Python, SpeechRecognition, NLP*

- Built a voice-activated personal assistant using Python NLP libraries to parse and execute system commands, improving command recognition accuracy by 20% through robust error handling.

EDUCATION

B.Tech in Computer Science and Engineering **2017 – 2021**

Radhakrishna Institute of Technology and Engineering, Bhubaneswar, Odisha | CGPA: 7.94 / 10 (74.37%)

Higher Secondary (+2), Science **2014 – 2017**

Stewart Science College, Cuttack, Odisha | Percentage: 52%

Matriculation **2013 – 2014**

S.S.V.M Tulasipur, Cuttack, Odisha | Percentage: 57%

CERTIFICATIONS & TRAINING

- Data Analysis Using Python – IBM
- Python (Basic) – HackerRank (Certificate ID: 0e618ad01c64)
- SQL (Basic) – HackerRank (Certificate ID: f2d8e83dae15)
- Python with Django – Central Tool Room and Training Centre (CTTC), Bhubaneswar

ADDITIONAL INFORMATION

- Available for immediate joining.
- Open to remote, hybrid, and on-site opportunities.
- Strong problem-solving ability, attention to detail, and communication skills.

LANGUAGES

English (Professional Proficiency), Hindi (Fluent), Odia (Native)